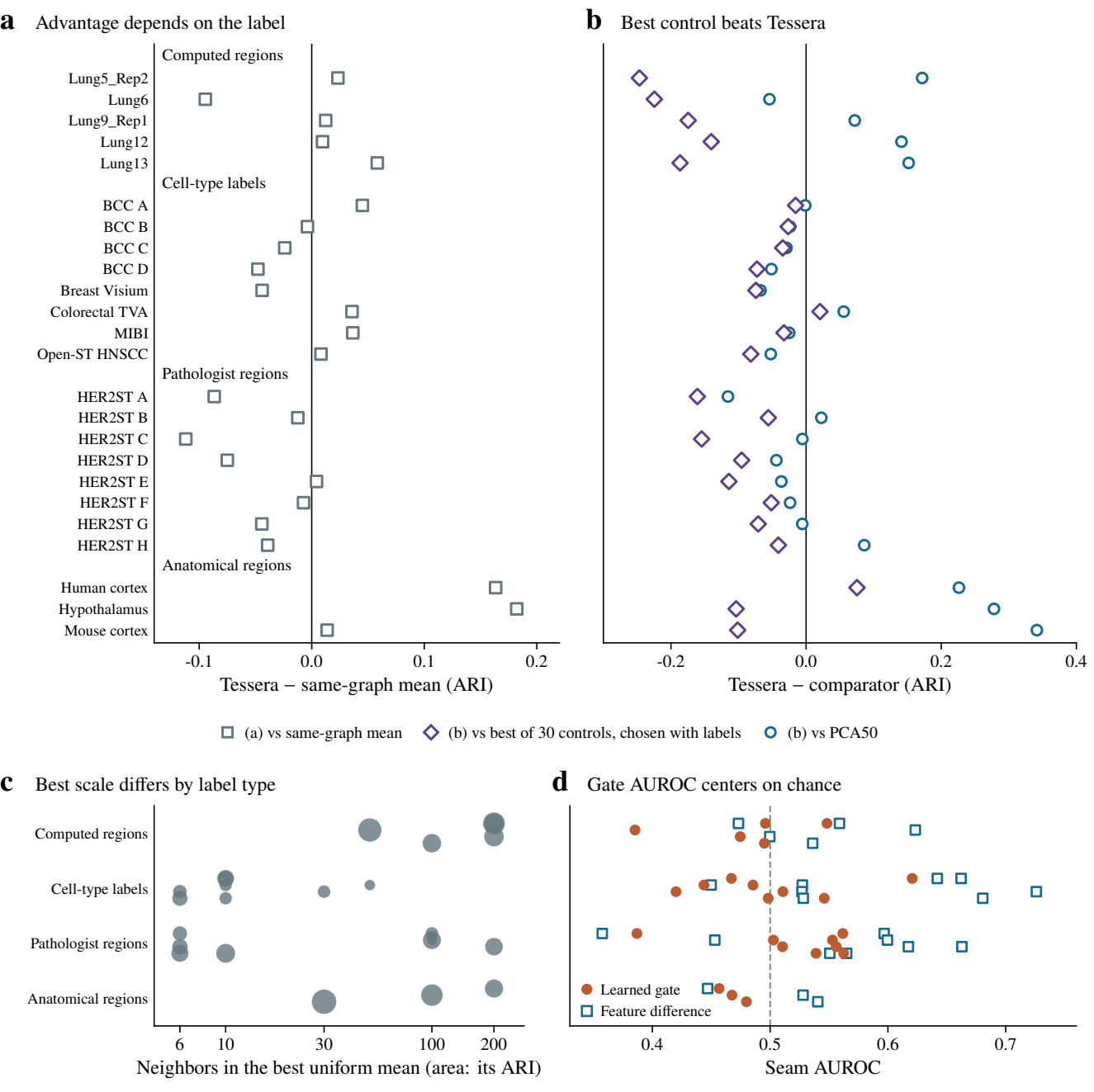

a
Advantage depends on the label
Computed regions
Lung5_Rep2
Lung6
Lung9_Rep1
Lung12
Lung13
Cell-type labels
BCC A
BCC B
BCC C
BCC D
Breast Visium
Colorectal TVA
MIBI
Open-ST HNSCC
Pathologist regions
HER2ST A
HER2ST B
HER2ST C
HER2ST D
HER2ST E
HER2ST F
HER2ST G
HER2ST H
Anatomical regions
Human cortex
Hypothalamus
Mouse cortex
-0.1
0.0
0.1
0.2
Tessera − same-graph mean (ARI)
b
Best control beats Tessera
-0.2
0.0
0.2
0.4
Tessera − comparator (ARI)
(a) vs same-graph mean
(b) vs best of 30 controls, chosen with labels
(b) vs PCA50
c
Best scale differs by label type
Computed regions
Cell-type labels
Pathologist regions
Anatomical regions
6
10
30
100
200
Neighbors in the best uniform mean (area: its ARI)
d
Gate AUROC centers on chance
Learned gate
Feature difference
0.4
0.5
0.6
0.7
Seam AUROC